Numeric Data

Chapter 3

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Outline

1. Regression
   - Goodness of Fit
   - Multicollinearity and Variance Inflation Factors
2. Nadaraya-Watson Kernel Regression
3. Dimensionality Reduction

1. Regression

Regression

- Regression is a statistical method used to analyze and model the relationship between a
  - response variable (also known as the dependent variable, typically denoted as Y)
  - one or more covariates (also known as independent variables, predictors, or features, typically denoted as X)
- to predict missing or unavailable numerical values
- primary objective is to identify the "best" mathematical function that fits the relationship between attributes
- a maps f from input X to output Y

$$\mathbf{y} = \mathbf{f}(\mathbf{x}) + \epsilon \tag{1}$$

- where ϵ is a random error term

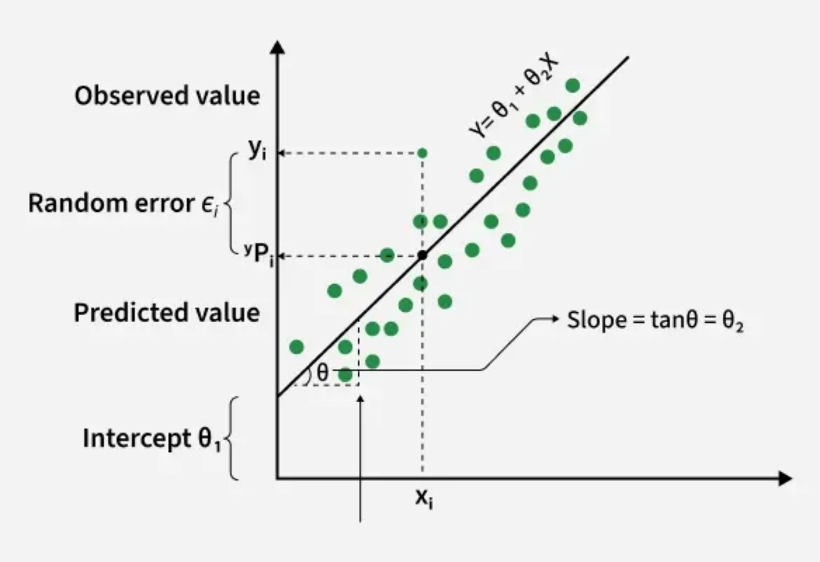
Univariate Linear Regression

- Univariate (or simple) linear regression involves exactly two attributes: one independent attribute (x) and one continuous output variable (y)
- model assumes a linear relationship expressed as:

$$\hat{y} = \theta x + \theta_0 + \epsilon$$

- where:
    - $\hat{y}$: response variable
    - x : predictor variable
    - θ : regression coefficient
        - weight (or slope) representing the importance of the attribute
    - θ_0 bias scalar (or y-intercept)
    - ϵ : random error

Univariate Linear Regression



Example

Land price in Lamachaur

Land area (aana)	Price (crore)
4.12	0.715
5.1	0.72
5.2	0.82
8.18	1.15
16.95	2.83
19.61	2.95
⋮	⋮

Example

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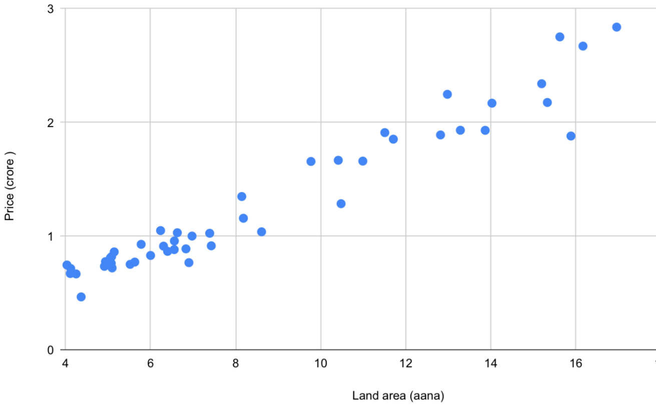

How can we learn to predict the prices of other land in Lamachaur?

Example

Land price in Lamachaur

Land area (aana)	Price (crore)
4.12	0.715
5.1	0.72
5.2	0.82
8.18	1.15
16.95	2.83
19.61	2.95
$\vdots$	$\vdots$

- x^i: **input** variables (land area)
- y^i: **output** variable :to predict (price)
- m: **training example**
- $\{(x^i, y^i);\ i = 1, ..., m\}$: **training set**
- $\mathcal{X} = (x^1,\ x^2, \cdots,\ x^i,\ \cdots,\ x^m)$: space of input values
- $\mathcal{Y} = (y^1,\ \cdots,\ y^i,\ \cdots,\ y^m)$:space of output values
- $\mathcal{X} = \mathcal{Y} = \mathbb{R}$

$$\begin{array}{c|c} x^2 & 5.1 \\ y^2 & 0.72 \end{array}$$

Simple Linear Regression & OLS Estimation

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \text{where } \varepsilon_i \sim iid(0, \sigma^2)$$

Find the straight line that best predicts the response variable.

- Y_i : Actual (observed) value
- $\hat{Y}_i$: Predicted value from the regression line
- $\bar{Y}$: Mean (average) of all observations

For every observation: $Y_i - \bar{Y} = (Y_i - \hat{Y}) + (\hat{Y}_i - \bar{Y})$

$\boxed{SST = SSR + SSE}$ **SST - Total Sum of Squares:**

$$SST = \sum (Y_i - \bar{Y})^2$$

How much the observed data vary around the average value.

SSR - Regression Sum of Squares

$$SSR = \sum(\hat{Y}_i - \bar{Y})^2$$

- How much of the variability is explained by the regression model
- Larger SSR means: better prediction, stronger relationship

SSE - Error Sum of Squares

$$SSE = \sum(Y_i - \hat{Y}_i)^2$$

- How much error remains after fitting the regression line
- Smaller SSE means: predictions are close to actual observations, better model

Ordinary Least Squares (OLS) Goal: Minimize the Sum of Squared Errors (**SSE**):

$$\min_{\theta_0,\theta_1} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \min_{\theta_0,\theta_1} \sum_{i=1}^{n} (Y_i - \theta_0 - \theta_1 X_i)^2$$

regression algorithm finds the line that makes this quantity as small as possible

OLS Estimators (Closed-Form): Slope ($\hat{\theta}_1$)

$$\hat{\theta}_1 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}$$

Intercept ($\hat{\theta}_0$)

$$\hat{\theta}_0 = \bar{Y} - \hat{\theta}_1 \bar{X}$$

Interpretation

- The regression line is the **best-fit line**.
- Residual = Actual value − Predicted value.
- The Least Squares Method chooses the line that minimizes

$$SSE = \sum (Y_i - \hat{Y}_i)^2.$$

- A smaller SSE indicates a better-fitting model.

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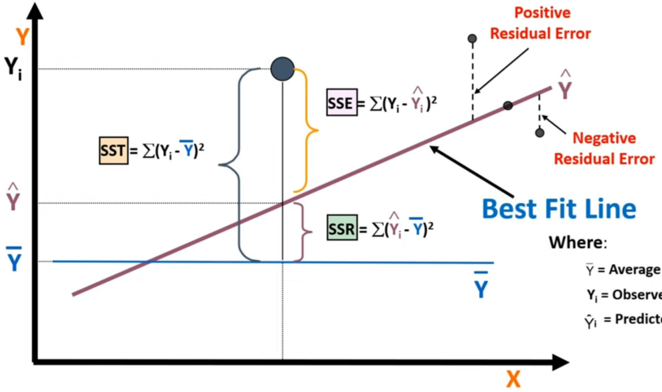
$$SSE = \sum (Y_i - \hat{Y}_i)^2.$$

- A smaller SSE indicates a better-fitting model.
  Problem: In the real world, outcomes are rarely driven by a single factor
  - Example: The price of a house isn't just determined by its size; it's
    - also driven by location, number of bedrooms, age, and local crime rates
  - Multiple regression allows us to capture this complexity



$$SST = \sum (Y_i - \bar{Y})^2$$

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2$$

$$SSE = \sum (Y_i - \hat{Y}_i)^2$$

Where:

- $\bar{Y}$ = **Average value of the dependent variable**
- Y_i = **Observed values of the dependent variable**
- $\hat{Y}_i$ = **Predicted value of Y for the given X_i value**

Goodness of Fit

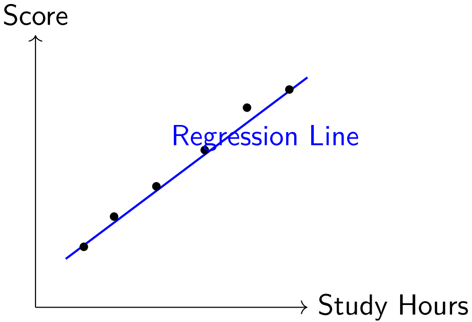
Goodness of Fit

After fitting a regression model, an important question is:

- How Well Does the Regression Model Fit the Data?

A good regression model should

- Predict values close to the actual observations.
- Explain most of the variation in the response variable.
- Produce small prediction errors.



A better model has predictions that are closer to the observed values

Coefficient of Determination (R^2)

measures the proportion of the total variation in the response variable that is explained by the regression model

The proportion of variability explained by the model is

$$R^2 = \frac{SSR}{SST}$$

Substitute $SSR = SST - SSE$ Therefore,

$$R^2 = \frac{SST - SSE}{SST} = 1 - \frac{SSE}{SST}$$

Interpretation

- $R^2 = 0:$ Model explains none of the variation (same as the mean)
- $R^2 = 1:$ Perfect fit (prediction)
  - Every data point lies exactly on the regression line
- $R^2 < 0:$ model makes predictions that are worse than simply using the mean

Multivariate Linear Regression

statistical technique used to predict a continuous dependent variable using two or more independent variables

$$Y = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \cdots + \theta_p X_p + \varepsilon = \sum_{i=1}^{p} \theta_i x_i + \theta_0$$

Where:

- Y : Dependent (response) variable
- $X_1, X_2, \ldots, X_p$: Independent (predictor) variables
- θ_0 : Intercept
- $\theta_1, \theta_2, \ldots, \theta_p$: Regression coefficients
- ε : Random error

Example

Predict a student's exam score using

$$\boxed{\begin{array}{c} X_1 = \text{Study Hours} \\ X_2 = \text{Attendance} \\ X_3 = \text{Sleep Hours} \end{array}} \Longrightarrow \boxed{Y = \text{Exam Score}}$$

Unlike Simple Linear Regression, $Y = \theta_0 + \theta_1 X$
Multiple Regression uses several predictors simultaneously:

$$Y = \theta_0 + \theta_1 X_1 + \theta_2 X_2 + \theta_3 X_3$$

Goal Estimate the coefficients that minimize the prediction error using the Least Squares Method.

Matrix Representation of Multiple Regression

For n observations and p predictors,

$$\mathbf{y} = \mathbf{X}\theta + \varepsilon$$

where

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}, \qquad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}$$

The Ordinary Least Squares (OLS) estimator is

$$\boxed{\hat{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T Y}$$

Interpretation

OLS finds the coefficient vector that minimizes the squared prediction errors.

Assumptions of Multiple Linear Regression

For reliable predictions, the following assumptions should hold.

1. Linear relationship $Y = \theta_0 + \theta_1 X_1 + \cdots + \theta_p X_p$
2. Independent observations
3. Normally distributed residuals
4. Constant variance of residuals (Homoscedasticity)
5. No multicollinearity $VIF < 5$: generally acceptable

If assumptions are violated

- Coefficients become unstable.
- Confidence intervals become unreliable.
- Predictions may be inaccurate.

Predicting Exam Score from Study Hours, Attendance, and Sleep Hours

Student	Study Hours	Attendance (%)	Sleep Hours	Score
1	2	70	6	55
2	3	75	7	58
3	4	80	6	68
4	5	85	7	74
5	6	90	8	82
6	2	65	5	50
7	3	70	6	54
8	4	75	7	68
9	5	80	7	72
10	6	90	8	84

Adding bias (all ones), the design matrix ($\mathbf{X}$) and $\mathbf{y}$ becomes

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 70 & 6 \\ 1 & 3 & 75 & 7 \\ 1 & 4 & 80 & 6 \\ 1 & 5 & 85 & 7 \\ 1 & 6 & 90 & 8 \\ 1 & 2 & 65 & 5 \\ 1 & 3 & 70 & 6 \\ 1 & 4 & 75 & 7 \\ 1 & 5 & 80 & 7 \\ 1 & 6 & 90 & 8 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 55 \\ 58 \\ 68 \\ 74 \\ 82 \\ 50 \\ 54 \\ 68 \\ 72 \\ 84 \end{bmatrix}$$

$$\hat{\theta} = \left[(X^T X)^{-1} X^T\right] Y = \begin{bmatrix} 2.378 \\ 4.527 \\ 0.559 \\ 0.363 \end{bmatrix}$$

Given the fitted regression model

$$\boxed{\hat{Y} = 2.378 + 4.527X_1 + 0.559X_2 + 0.363X_3}$$

- Study Hours $\theta_1 = 4.527$: One additional study hour increases the predicted score by 4.53 marks
- Attendance $\theta_2 = 0.559$: Every additional 1% attendance increases the score by 0.559 marks
- Sleep $\theta_3 = 0.363$: One extra hour of sleep increases the predicted score by 0.36

Predict the score of a student with

$$X_1 = 5, \qquad X_2 = 85, \qquad X_3 = 7$$

Substitute the values

$$\begin{aligned}\hat{Y} &= 2.378 + 4.527(5) + 0.559(85) + 0.363(7) \\ &= 2.378 + 22.635 + 47.515 + 2.541 = 75.069\end{aligned}$$

Interpretation: student is expected to score approximately 75 marks

Adjusted R^2

Problem: Adding more predictors always increases R^2

- OLS always chooses the coefficients that minimize the Sum of Squared Errors (SSE)
- model has more parameters, training R^2 often increases slightly
  - creates the false impression that the model is improving
    - phenomenon is called overfitting

Adjusted R^2 penalizes unnecessary variables.

$$\boxed{R^2_{adj} = 1 - \frac{SSE/(n-p-1)}{SST/(n-1)}} = 1 - \left(1 - R^2\right)\frac{n-1}{n-p-1}$$

where

- $n =$ Number of observations
- $p =$ Number of predictors

Penalty in Regression: Denominator of Error Term

Denominator: $n - p - 1$
Every time we add one predictor ($p \uparrow$), the denominator becomes smaller.

Example: $n = 100$

- Model A: $p = 2$ predictors Degrees of freedom: $100 - 2 - 1 = 97$
- Model B: $p = 20$ predictors Degrees of freedom: $100 - 20 - 1 = 79$

Observation:

$$79 < 97$$

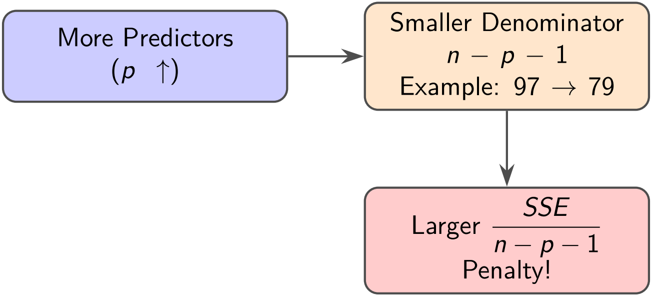
Since the denominator is smaller:

$$\frac{SSE}{79} > \frac{SSE}{97}$$

unless SSE decreases sufficiently.

Conclusion: This reduction in degrees of freedom is the **penalty** for adding predictors.

Penalty in Regression



Flow: More predictors $\Rightarrow$ fewer degrees of freedom $\Rightarrow$ larger error term denominator $\Rightarrow$ penalty.

Example: Used Car Price Prediction

- y: Price of the car (in $1,000s)
- x_1: Age of the car (years)
- x_2: Mileage (in 10,000s of miles)

You collect data on exactly 3 cars from a local dealership:

Car	**Age (x_1)**	**Mileage (x_2)**	**Price (y)**
A	2	2	18
B	4	5	12
C	5	4	11

Design Matrix and Target Vector:

$$X = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 4 & 5 \\ 1 & 5 & 4 \end{bmatrix}, \quad y = \begin{bmatrix} 18 \\ 12 \\ 11 \end{bmatrix}$$

Compute X^TX and X^Ty:

$$X^TX = \begin{bmatrix} 3 & 11 & 11 \\ 11 & 45 & 44 \\ 11 & 44 & 45 \end{bmatrix}, \quad X^Ty = \begin{bmatrix} 41 \\ 139 \\ 140 \end{bmatrix}$$

Inverse of (X^TX):

$$(X^TX)^{-1} = \frac{1}{25} \begin{bmatrix} 89 & -11 & -11 \\ -11 & 14 & -11 \\ -11 & -11 & 14 \end{bmatrix}$$

Estimated Coefficients:

$$\hat{\beta} = (X^TX)^{-1}X^Ty = \frac{1}{25} \begin{bmatrix} 89 & -11 & -11 \\ -11 & 14 & -11 \\ -11 & -11 & 14 \end{bmatrix} \begin{bmatrix} 41 \\ 139 \\ 140 \end{bmatrix} = \begin{bmatrix} 23.2 \\ -1.8 \\ -0.8 \end{bmatrix}$$

$$\beta_0 = 23.2, \quad \beta_1 = -1.8, \quad \beta_2 = -0.8$$

Regression Equation:

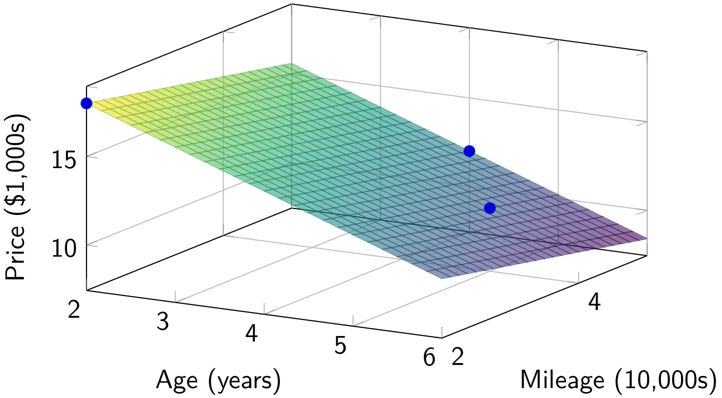
$$\hat{y} = 23.2 - 1.8x_1 - 0.8x_2$$

Interpretation:

- Intercept (23.2): A brand-new car (0 years, 0 mileage) costs about \$23,200.
- Age (-1.8): Each additional year reduces price by \$1,800.
- Mileage (-0.8): Each additional 10,000 miles reduces price by \$800.

Geometric: Regression Plane

Interpretation in 3D: The regression model fits a plane in the space of Age (x_1), Mileage (x_2), and Price (y).



plane minimizes squared distances to the observed points.

Calculation of R^2 and R^2_{adj} of car price

Step 1: Mean of y

$$\bar{y} = \frac{18 + 12 + 11}{3} = \frac{41}{3} \approx 13.67$$

Step 2: Total Sum of Squares (SST)

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \frac{86}{3} \approx 28.67$$

Step 3: Error Sum of Squares (SSE)

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = 0$$

Calculation of R^2 and R^2_{adj} of car price

Residuals Table:

Car	Actual (y_i)	Predicted ($\hat{y}_i$)	Residual ($y_i - \hat{y}_i$)
A	18	$23.2 - 1.8(2) - 0.8(2) = 18$	0
B	12	$23.2 - 1.8(4) - 0.8(5) = 12$	0
C	11	$23.2 - 1.8(5) - 0.8(4) = 11$	0

Step 4: R^2

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{0}{86/3} = 1.0$$

Interpretation: The model explains 100% of the variance in car prices. With 3 data points and 3 parameters, the regression plane passes exactly through every point.

Calculation of R^2 and R^2_{adj} of car price

Step 5: Calculate Adjusted R^2_{adj}

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{n-1}{n-p-1} = 1 - (1-1)\frac{3-1}{3-2-1} = 1 - \frac{0}{0} \Rightarrow \text{Undefined}$$

Reason:

- Residual degrees of freedom: $df_{\text{res}} = n - p - 1 = 3 - 2 - 1 = 0$
- With $df_{\text{res}} = 0$, we cannot estimate the error variance σ^2.
- Division by zero occurs in the formula, making Adjusted R^2 undefined.

Interpretation: With 3 data points and 3 parameters, the regression plane passes exactly through every point. There are no residual degrees of freedom left, so Adjusted R^2 cannot be computed.

Multicollinearity and Variance Inflation Factors

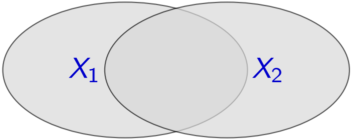
Multicollinearity and Variance Inflation Factors (VIF)

- **Multicollinearity** occurs when two or more independent variables (predictors) in a multiple regression model are highly correlated with each other
- If x_1 and x_2 are highly correlated, they contain overlapping information
  - model struggles to determine the individual effect of x_1 on y separate from the effect of x_2
- one independent variable can be predicted from the others with a high degree of accuracy

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n + \varepsilon$$

- model cannot determine θ_1 and θ_2 value
- variances explode!
- p-values become useless

Multicollinearity



Overlapping Information (High Correlation)

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n + \varepsilon$$

Example: housing price prediction

- $\hat{y}$ price of house
- x_1 size of house, x_2 Number of rooms, and so on
- Larger house: tends to have more rooms
- model cannot measure the influence of each variable
- regression model for a prediction: multicollinearity is less critical
- regression model to assess the influence of the independent variable: No multicollinearity

Coefficient of Determination R^2

Determine one variable influenced by other variables or a combination of variables
create auxiliary regression model

$$\hat{x}_1 = \theta_0 + \theta_2 x_2 + \cdots + \theta_n x_n + \varepsilon$$
$$\hat{x}_2 = \theta_0 + \theta_1 x_1 + \cdots + \theta_n x_n + \varepsilon$$
$$\hat{x}_k = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \varepsilon$$

- x_1 can predict accurately using other independent variables
  - x_1 becomes unnecessary
    - its information is already caputared by other variables

Method to detect multicolinearity

- R^2 explains how well the independent variables
  - explain the variability of the dependent variables

Coefficient of Determination R^2

- High $R^2 \Rightarrow x_1$: highly correlated with the other independent variables
  - sign of multicolinearity
- Using R^2, calculate
  1. Tolerance $T = 1 - R^2$
  2. Variance Inflation Factor $VIF = \frac{1}{1-R^2}$
- If $T < 0.1$ indicates potential multicolinearity, and caution is required
- $VIF > 10$: warning sign of multicolinearity requires further investigation

The VIF Formula:

$$VIF_j = \frac{1}{1 - R_j^2}$$

- If X_j is completely independent, $R_j^2 = 0 \implies VIF = 1$
- If X_j is perfectly predicted by others, $R_j^2 \to 1 \implies VIF \to \infty$

Variance Inflation Factor (VIF)

Let's look at the variance of a specific coefficient $\hat{\beta}_j$:

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2} \times \frac{1}{1 - R_j^2}$$

- σ^2: Error variance.
- $\sum(x_{ij} - \bar{x}_j)^2$: Total variation in X_j (SST_j).
- R_j^2: The R^2 from regressing X_j on *all other predictors*
  - If $R_j^2 \to 1 \implies VIF \to \infty$. The variance is literally *inflated* by this factor!

Note: *R_j^2 is NOT the R-squared of the main model predicting Y. It is the R-squared of an auxiliary model predicting X_j*

Variance Inflation Factor (VIF)

Table: Interpretation of VIF

VIF Value	Interpretation
1	No multicollinearity
1 – 5	Moderate correlation, acceptable
5 – 10	High correlation, potential problem
> 10	Serious multicollinearity issue

Note: VIF quantifies how much the variance of a regression coefficient is inflated due to multicollinearity. Lower values are better; values above 10 indicate severe issues.

Variance Inflation Factor (VIF)

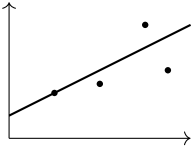
Solution

1. Remove the independent correlated variable with less significance
2. Combined with other correlated variables
3. Use dimensionality reduction techniques like Principal Component Analysis (PCA)

Parametric vs. Non-Parametric Regression

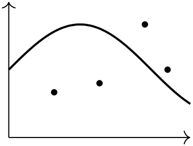
Parametric (e.g., OLS)

- Assumes a strict shape (e.g., a straight line).
- Fast, highly interpretable.
- Fails if the true relationship is complex/non-linear.



Non-Parametric (e.g., NW)

- "Let the data speak for itself."
- No strict global shape assumed.
- Fails if data is too sparse (curse of dimensionality).



Misconception: does not mean there are zero parameters. Rather, parameters are not fixed in advance and can grow as you get more data.

Parametric vs. Non-Parametric Regression

Feature	Parametric Regression	Non-Parametric Regression
Assumption	Assumes a specific functional form (e.g., $y = \beta_0 + \beta_1 x$)	Makes no strict assumptions about the functional form
Flexibility	Rigid. If the true relationship is complex, a line will fail	Highly flexible. Can adapt to almost any shape in the data
Data Needs	Works well with small datasets	Requires large datasets to accurately map the underlying function
Interpretability	High. Coefficients explain the exact effect of each variable	Low. Acts more like a "black box" (especially tree-based models)

Motivation: Why Linear Regression is Not Enough?

Linear regression assumes one straight line fits all data.

$$y = \beta_0 + \beta_1 x$$

- Real-world data are often curved.
- Growth slows over time.
- Relationships may be seasonal or nonlinear.

But many real-world relationships are:

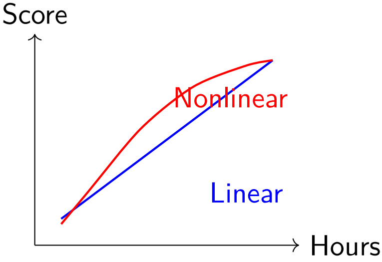
- Curved
- Wavy
- Seasonal
- Nonlinear

Examples:

- Temperature vs. electricity demand
- Age vs. income
- Blood pressure vs. age
- AQI vs. respiratory diseases

Linear vs Nonlinear Relationship

- Linear regression is powerful but limited
- Nonlinear methods (e.g., kernel regression, splines, tree-based models) can capture complex patterns in data



2. Nadaraya-Watson Kernel Regression

What is Kernel Regression?

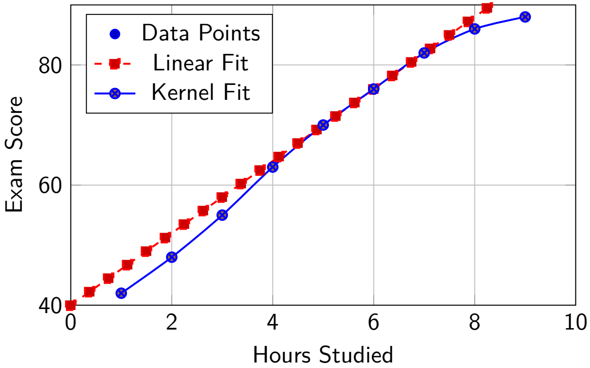
- non-parametric regression technique that estimates the output value by averaging nearby observations
- Instead of fitting one global equation, Kernel Regression fits the curve locally around each prediction point

Think of it as:
"To predict today's value, give more importance to nearby observations and less importance to distant observations."
Closer points receive larger weights than distant points.

Exam Score Example

Reality: Improvement slows after 6 hours. A straight line cannot perfectly describe this pattern.

Kernel Regression: Provides a flexible curve that adapts to the data.



Kernel regression captures the diminishing returns after 6 hours, while linear regression cannot.

What is a Kernel?

A kernel is a weighting function.

$$\text{Nearby observations} \Rightarrow \text{High weight}$$
$$\text{Far observations} \Rightarrow \text{Low weight}$$

K-Nearest Neighbors (KNN) regression asks, "What is the average of the K closest points?"
Nadaraya-Watson asks, "What is the weighted average of ALL points, where the weights are determined by how close they are?"

Distance	Weight	Importance
0	1.00	Very High
1	0.80	High
2	0.50	Medium
4	0.10	Low

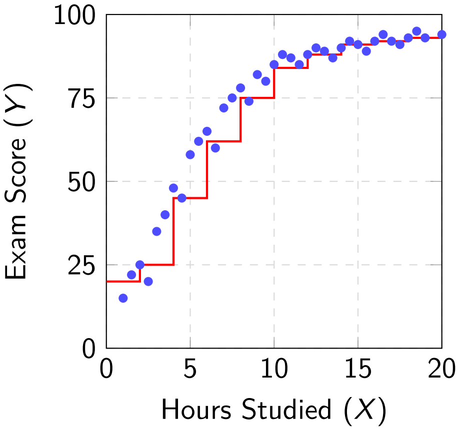
The Problem Setup

- **Goal:** Predict a student's final exam score (Y) based on hours studied (X).
- **The Reality:** Studying 10 hours gives a massive boost, but studying 20 hours gives a tiny boost (fatigue sets in).
- **The Challenge:** The relationship is non-linear. How do we model it?

Approach: K-Nearest Neighbors (KNN)

The Jagged Steps

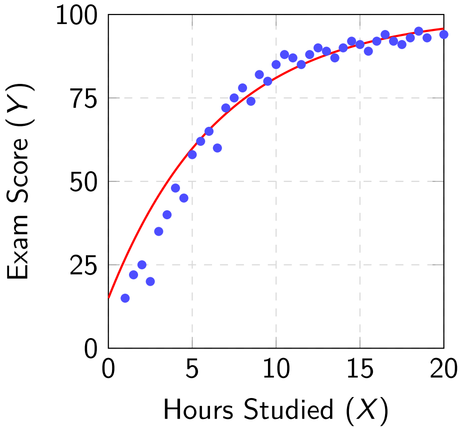
- Averages the K closest points (e.g., $K = 3$).
- **Pros:** Captures non-linearity.
- **Cons:** Creates a blocky, discontinuous step function. Awkward jumps!



Approach: Nadaraya-Watson (NW)

The Smooth Curve (The "Goldilocks" Solution)

- Takes a **weighted average** of *all* points.
- Weights decay smoothly with distance (the Kernel).
- **Result:** A beautifully smooth, continuous curve that flows naturally with the data!



Nadaraya-Watson Kernel Regression

- non-parametric method that predicts a value by taking a weighted average of nearby data points
- uses a **Kernel** to determine how weights decay with distance, and a **Bandwidth** to control the size of the neighborhood
- trades the rigid assumptions of linear regression for the smooth, data-driven flexibility of a localized weighted average
- draws a beautifully smooth curve that flows naturally through the data

Nadaraya-Watson Kernel Regression

- **No rigid assumptions:** Doesn't force a straight line.
- **No jagged jumps:** Smoother than KNN.
- **The Trade-off:** Highly flexible and smooth, but computationally heavier and requires tuning the **Bandwidth** (h), which controls the size of the spotlight.

Nadaraya-Watson Kernel Regression

Estimator:

$$\hat{m}(x) = \frac{\sum_{i=1}^{n} K_h(x - x_i)\, y_i}{\sum_{i=1}^{n} K_h(x - x_i)}$$

Components:

- $\hat{m}(x)$: Predicted value (estimated conditional expectation $E[Y|X = x]$).
- y_i: Observed target value for the i-th data point.
- x_i: Observed feature value for the i-th data point.
- $K_h(x - x_i)$: Kernel weight based on distance from x.

Why the denominator?

- Ensures all weights sum to 1.
- Converts the numerator into a proper weighted average.
- Provides a mathematically sound estimate of $E[Y|X = x]$.

Nadaraya-Watson Kernel Regression

Kernel Function:

$$K_h(x - x_i) = \frac{1}{h} K\left(\frac{x - x_i}{h}\right)$$

Gaussian Kernel:

$$K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^2\right)$$

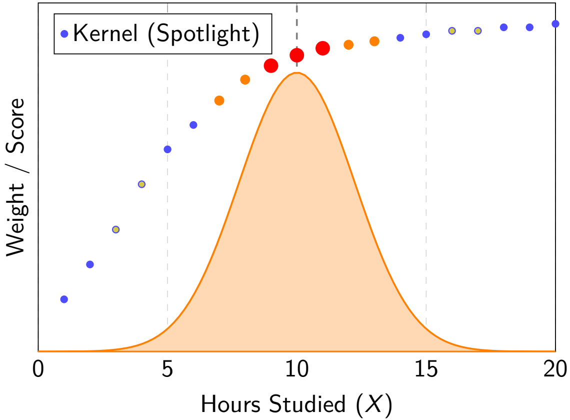
Nadaraya–Watson Estimator:

$$\hat{m}(x) = \frac{\sum_{i=1}^{n} \frac{1}{h} K\left(\frac{x-x_i}{h}\right) y_i}{\sum_{i=1}^{n} \frac{1}{h} K\left(\frac{x-x_i}{h}\right)}$$

Interpretation:

- The kernel $K_h(\cdot)$ assigns higher weights to points closer to x.
- The denominator normalizes the weights so they sum to 1.
- The estimator computes a smooth, locally weighted average of y_i values.

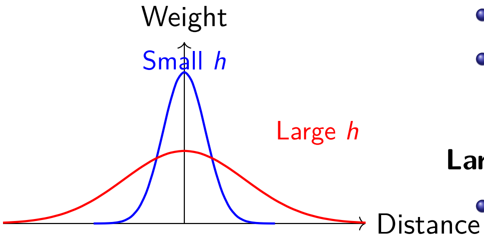
Visualizing the Kernel and Bandwidth



- **Red/Large:** Close to target → High weight.
- **Orange/Medium:** Further away → Lower weight.
- **Yellow/Small:** Far away → Almost zero weight.

Role of Bandwidth (h)

The bandwidth h controls the width of the kernel $K_h(u) = \frac{1}{h}K(\frac{u}{h})$. It is the **most important hyperparameter**.



Small h (Under-smoothing):

- Only very close points matter.
- High Variance, Low Bias (Overfitting, wiggly curve).

Large h (Over-smoothing):

- Distant points get equal weight.
- Low Variance, High Bias (Underfitting, flat curve).

Note: Use the Goldilocks analogy. Small h is too picky, large h is too inclusive. We want the "just right" bandwidth that minimizes Mean Squared Error (MSE).

Numerical

Given dataset $X = [1, 2, 3, 4, 5]$, $Y = [2, 4, 1, 5, 3]$. Estimate $\hat{y}$ at $x = 2.5$ using Gaussian kernel with bandwidth $h = 1$.

Estimator:

$$\hat{y}(x) = \frac{\sum_{i=1}^{n} K_h(x - X_i) Y_i}{\sum_{i=1}^{n} K_h(x - X_i)}, \quad K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$$

Step 1: Distances $d = |x - x_i| \quad |2.5 - 1| = 1.5$

$$d = [1.5,\ 0.5,\ 0.5,\ 1.5,\ 2.5]$$

Step 2: Kernel Weights $K_h(u) = \exp\left(-\frac{u^2}{2h^2}\right)$; $u = |x - x_i|$

$$w_1 = \exp\left(-\frac{1.5^2}{2}\right) \approx 0.3247 \quad w_2 = \exp\left(-\frac{0.5^2}{2}\right) \approx 0.8825$$

$$\vdots \qquad \vdots$$

$$w = [0.3247,\ 0.8825,\ 0.8825,\ 0.3247,\ 0.0440]$$

Step 3: Weighted Sums

$$\text{Numerator} = \sum w_i Y_i = 6.8174, \quad \text{Denominator} = \sum w_i = 2.4584$$

Step 4: Prediction

$$\hat{y}(2.5) = \frac{6.8174}{2.4584} \approx 2.772$$

At $x = 2.5$, the predicted exam score is approximately **2.77**.

3. Dimensionality Reduction

Dimensionality Reduction

A technique to reduce the number of random variables under consideration by obtaining a set of principal variables.

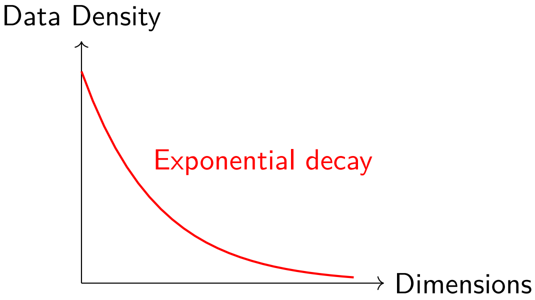
- High-dimensional data: datasets with many features/variables
- Goal: Transform data from high-dimensional space to lower-dimensional space
- Preserve important information while discarding noise and redundancy
- Two main approaches:
  - Feature Selection
  - Feature Extraction (PCA)

Curse of Dimensionality

Curse of Dimensionality

As the number of dimensions increases, the volume of the space increases exponentially, making data sparse.

- Distance metrics become less meaningful
- Data becomes sparse
- More data needed exponentially
- Overfitting risk increases
- Computational cost grows



Example

In 100 dimensions, you need $\approx 10^{100}$ data points to maintain density!

4. Why Feature Reduction?

Why Feature Reduction?

High Dimensions $\xrightarrow{\text{Feature Reduction}}$ Low Dimensions

Benefits:
Faster, Simpler, Better

1. **Computational Efficiency**: Faster training and prediction
2. **Storage Reduction**: Less memory and disk space
3. **Improved Performance**: Better generalization, less overfitting
4. **Visualization**: Plot data in 2D/3D
5. **Noise Reduction**: Remove irrelevant features

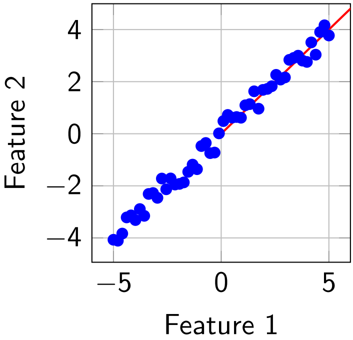
Real-life Examples

- **Image Processing**
  - 1000×1000 pixel image = 1,000,000 dimensions
  - Reduce to 100-500 principal components
- **Genomics**
  - Gene expression data: 20,000+ genes
  - Reduce to identify key patterns
- **Finance**
  - Portfolio optimization with 1000+ assets
  - Identify underlying risk factors
- **Natural Language Processing**
  - Word embeddings: 300-1000 dimensions
  - Reduce for visualization and efficiency

Feature Correlation

Features are often correlated, containing redundant information.

- Highly correlated features provide similar information
- Example: Height in cm vs Height in inches
- Correlation coefficient: $r \approx 1$ or $r \approx -1$
- Wastes computational resources
- Can cause multicollinearity in models



Correlation matrix helps identify redundant features

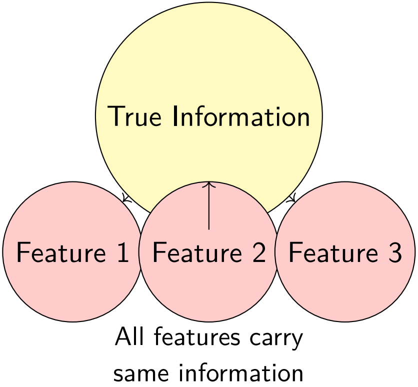
5. Redundant Information

Redundant Information

Redundancy: Multiple features encoding the same or similar information.

Examples:

- Temperature in °C and °F
- Date and Day of Week
- Multiple sensors measuring same quantity
- Derived features (total = sum of parts)



Solution
Keep only the essential information; remove redundancy through dimensionality reduction.

Principal Component Analysis

A statistical procedure that transforms correlated variables into uncorrelated principal components.

- Find directions of maximum variance in data
- Project data onto lower-dimensional subspace
- Preserve as much variance (information) as possible
- Create uncorrelated new features (principal components)

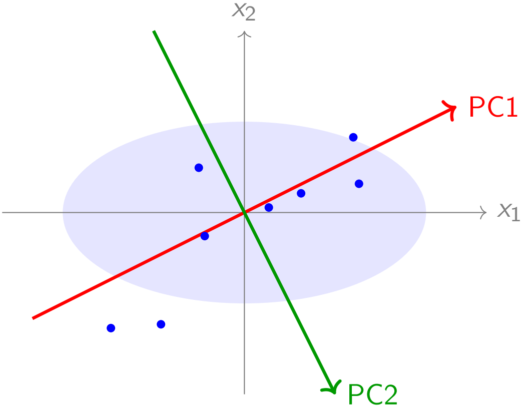
Intuition

If you could only keep one number to summarize each data point, which combination of features would tell you the most?

- PCA finds this optimal combination automatically

Intuitive Geometry of PCA

- Imagine data points in 2D space
- **PC1**: Direction of maximum variance (best fit line)
- **PC2**: Orthogonal to PC1, second most variance
- Rotate coordinate system to align with PCs
- Project points onto new axes



Geometric Interpretation

PCA finds the best viewing angle to see the data's structure with minimal information loss.

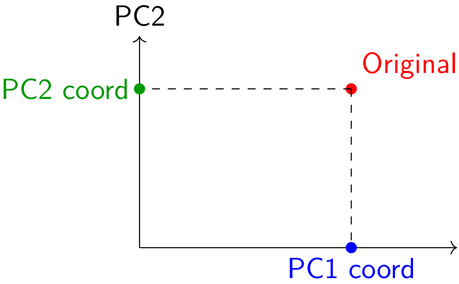
Intuitive Geometry of PCA

Variance Maximization

- PC1 captures the direction where data varies most
- PC2 captures the next most variance (orthogonal to PC1)
- Each subsequent PC is orthogonal to all previous ones

Projection:

- Drop perpendiculars from points to PC axes
- New coordinates = distances along PCs
- Keep top k PCs for k-dimensional representation



By keeping only PC1, we reduce 2D to 1D while preserving maximum variance

Behind PCA

PCA is not simply rotating the axes.

Objective: $\boxed{\text{Find a new coordinate system that maximizes data variance}}$
subject to $\|\mathbf{w}\| = 1$
where

- $\mathbf{w}$ = projection direction
- projected data = $\mathbf{Xw}$
- maximize variance of projected data

Optimization Problem
Find

$$\mathbf{w} = \arg\max Var(\mathbf{Xw})$$

subject to

$$\mathbf{w}^T\mathbf{w} = 1$$

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be the data matrix (n samples, p features) **Step 1 : Standardization**

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

where

- rows = observations
- columns = variables

Each feature is standardized

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}$$

where

- μ_j = feature mean
- σ_j = feature standard deviation

Without scaling, variables having larger units dominate PCA.

Step 2 : Covariance Matrix

PCA studies how variables vary together

Covariance matrix

$$S = \frac{1}{n-1}\mathbf{Z}^T\mathbf{Z} \qquad S = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \cdots \\ Cov(X_2, X_1) & Var(X_2) & \\ \vdots & & \ddots \end{bmatrix}$$

where

$$Cov(X, Y) = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$$

- Positive covariance $\uparrow X \Rightarrow \uparrow Y$
- Negative covariance $\uparrow X \Rightarrow \downarrow Y$
- If $C_{ij} = 0$, features i and j are uncorrelated
- If $C_{ij} > 0$, they vary together
- If $C_{ij} < 0$, they vary inversely

Step 3 : Eigenvalues and Eigenvectors

$$C\mathbf{v} = \lambda \mathbf{v}$$

where

- $\mathbf{v}$ = Eigenvector = Principal Component Direction
  - orthogonal (uncorrelated)
- λ = Eigenvalue = Variance Along That Direction
  - Eigenvalues rank PCs by importance (higher λ = more variance)

$$\lambda_1 > \lambda_2 > \cdots > \lambda_p \Rightarrow PC_1 > PC_2 > \cdots > PC_p$$

Explained Variance Ratio:

$$\text{Explained Variance} = \frac{\lambda_i}{\sum \lambda}$$

Step 4 : Transform the Data into the new subspace

Collect eigenvectors

$$\mathbf{W} = [v_1, v_2, \cdots, v_p]$$

Project data

$$\mathbf{X}_{PCA} = \mathbf{ZW}$$

Numerical

Student scores

Student	*Math*	*Physics*
A	80	82
B	85	88
C	90	91
D	95	96

Numerical

Student scores

Student	*Math*	*Physics*
A	80	82
B	85	88
C	90	91
D	95	96

- Number of samples: $n = 4$
- Number of features: $d = 2$
- **Goal:** Apply PCA to find the principal components and reduce dimensionality if possible.

Step 1: Center the Data

Mean Centering: Subtract the mean of each feature from the data points to center the data at the origin.

- not full Standardization (Z-score normalization).
- "Math" and "Physics" were measured on the exact same scale (test scores out of 100)

Calculate Means:

$$\mu_{\text{Math}} = \frac{80 + 85 + 90 + 95}{4} = 87.5$$
$$\mu_{\text{Physics}} = \frac{82 + 88 + 91 + 96}{4} = 89.25$$

Centered Data Matrix ($\mathbf{X}_c$):

$$\mathbf{X}_c = \begin{bmatrix} -7.5 & -7.25 \\ -2.5 & -1.25 \\ 2.5 & 1.75 \\ 7.5 & 6.75 \end{bmatrix}$$

Step 2: Compute the Covariance Matrix $\mathbf{C} = \frac{1}{n-1}\mathbf{X}_c^T\mathbf{X}_c$, where $n = 4$

Calculations:

$$\mathbf{C} = \frac{1}{3}\begin{bmatrix} -7.5 & -2.5 & 2.5 & 7.5 \\ -7.25 & -1.25 & 1.75 & 6.75 \end{bmatrix}\begin{bmatrix} -7.5 & -7.25 \\ -2.5 & -1.25 \\ 2.5 & 1.75 \\ 7.5 & 6.75 \end{bmatrix} = \begin{bmatrix} 41.667 & 37.500 \\ 37.500 & 34.250 \end{bmatrix}$$

Step 3: Calculate Eigenvalues Find λ by solving $\det(\mathbf{C} - \lambda\mathbf{I}) = 0$.

$$\begin{vmatrix} 41.667 - \lambda & 37.500 \\ 37.500 & 34.250 - \lambda \end{vmatrix} = 0$$

$$(41.667 - \lambda)(34.250 - \lambda) - (37.500)^2 = 0$$
$$\lambda^2 - 75.917\lambda + 20.833 = 0$$

- $\lambda_1 \approx \mathbf{75.641}$ (Captures the most variance)
- $\lambda_2 \approx \mathbf{0.275}$ (Captures the least variance)

Solve $(\mathbf{C} - \lambda\mathbf{I})\vec{v} = 0$ for each eigenvector $(v_1,\ v_2)$, then normalize to unit vector $(u_1,\ u_2)$

For $\lambda_1 = 75.641$**:** find $v_1,\ u_1$

$$C - \lambda I = \begin{bmatrix} 41.667 - 75.641 & 37.500 \\ 37.500 & 34.250 - 75.641 \end{bmatrix} = \begin{bmatrix} -33.974 & 37.500 \\ 37.500 & -41.391 \end{bmatrix}$$

$$\begin{bmatrix} -33.974 & 37.500 \\ 37.500 & -41.391 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Equation from first row:

$$-33.974x + 37.500y = 0 \quad \Rightarrow \quad y \approx 0.906x$$

Choose $x = 1$, then $y = 0.906$ and Unnormalized eigenvector:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0.906 \end{bmatrix}$$

Normalize

$$\|\vec{v}_1\| = \sqrt{1^2 + 0.906^2} \approx 1.349$$

Normalized eigenvector for $\lambda_1 = 75.641$ is:

$$\vec{u}_1 = \frac{1}{1.349}\begin{bmatrix} 1 \\ 0.906 \end{bmatrix} \approx \begin{bmatrix} 0.741 \\ 0.671 \end{bmatrix}$$

For $\lambda_2 = 0.275$: find $v_1, \ u_1$

$$C - \lambda I = \begin{bmatrix} 41.667 - 0.275 & 37.500 \\ 37.500 & 34.250 - 0.275 \end{bmatrix} = \begin{bmatrix} 41.392 & 37.500 \\ 37.500 & 33.975 \end{bmatrix}$$

$$41.392x + 37.500y = 0 \quad \Rightarrow \quad y = -\frac{41.392}{37.500}x \approx -1.104x$$

Choose $x = 1$, then $y = -1.104$ and unnormalized eigenvector:

$$\vec{v}_2 = \begin{bmatrix} 1 \\ -1.104 \end{bmatrix}$$

Normalize

$$\|\vec{v}_2\| = \sqrt{1^2 + (-1.104)^2} \approx 1.490$$

Normalized eigenvector for $\lambda_2 = 0.275$ is:

$$\vec{u}_2 = \frac{1}{1.490}\begin{bmatrix} 1 \\ -1.104 \end{bmatrix} \approx \begin{bmatrix} 0.671 \\ -0.741 \end{bmatrix}$$

Tips: covariance matrix is symmetric, its eigenvectors are strictly orthogonal (perpendicular)

- find $\vec{v}_2$ instantly by taking $\vec{v}_1 = \begin{bmatrix} 1 \\ 0.906 \end{bmatrix}$, swapping the coordinates, and negating one of them: $\vec{v}_1 = \begin{bmatrix} -0.906 \\ 1 \end{bmatrix}$. divide both by -0.906, get exactly $\begin{bmatrix} 1 \\ -1.104 \end{bmatrix}$

Step 4: Explained Variance

Total Variance: $\lambda_1 + \lambda_2 = 75.641 + 0.275 = 75.916$

Explained Variance Ratio (EVR):

- **PC1 EVR:** $\frac{75.641}{75.916} \approx \mathbf{99.64}\%$ and **PC2 EVR:** $\frac{0.275}{75.916} \approx \mathbf{0.36}\%$

first principal component alone captures almost all (99.64%) of the information in the dataset

Step 5: Project the Data

Transform the centered data into the new subspace by multiplying $\mathbf{X}_c$ by the eigenvector matrix $\mathbf{U} = [\vec{u}_1, \vec{u}_2]$

$$\mathbf{Z} = \mathbf{X}_c\mathbf{U} = \begin{bmatrix} -7.5 & -7.25 \\ -2.5 & -1.25 \\ 2.5 & 1.75 \\ 7.5 & 6.75 \end{bmatrix} \begin{bmatrix} 0.741 & 0.671 \\ 0.671 & -0.741 \end{bmatrix}$$

Projected Data Matrix (Z):

$$\mathbf{Z} = \begin{bmatrix} -10.42 & 0.34 \\ -2.69 & -0.75 \\ 3.03 & 0.38 \\ 10.09 & 0.03 \end{bmatrix}$$

Interpretation of the Results

PC1: Overall Ability

- **Variance:** 99.64%
- **Weights:** Math (0.741), Physics (0.671)
- **Meaning:** Represents a student's general academic performance. Both subjects contribute positively and almost equally

PC2: Subject Difference

- **Variance:** 0.36%
- **Weights:** Math (0.671), Physics (-0.741)
- **Meaning:** variance is so low, indicates that there is almost no variation in "relative strength" among these students
    - if a student is good at Math, they are proportionally just as good at Physics

Conclusion

Final Verdict

For this specific dataset, PCA reveals that a single dimension (PC1) is sufficient to capture virtually all the differences between the students.

- reduce this 2D dataset to 1D with almost zero loss of information.
- new 1D feature (PC1) effectively acts as an "Overall Score" for each student.
- demonstrates the power of PCA in identifying and eliminating redundant information in correlated features.